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Masaya Arai, PhD

Curriculum Vitae

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Education

2026	Ph.D., Graduate School of Medicine, University of Osaka
2019	M.Sc., Graduate School of Agriculture, Kobe University
2017	B.A., Faculty of Agriculture, Kobe University

Positions

2026-Present	Postdoctoral Associate, The Rockefeller University Advisor: Daniel Mucida
2025-2026	Visiting Student, The Rockefeller University Advisor: Daniel Mucida
2023-2025	Specially Appointed Researcher, Immunology Frontier Research Center, University of Osaka Advisor: Shimon Sakaguchi (University of Osaka)
2020-2023	JSPS Research Fellow, Immunology Frontier Research Center, University of Osaka
2020-2023	Research Assistant, Immunology Frontier Research Center, University of Osaka
2019-2023	PhD program, Graduate School of Medicine, University of Osaka Advisor: Shimon Sakaguchi (University of Osaka) Thesis committee: Kiyoshi Takeda, Kazuyo Moro, Masahiro Yamamoto (University of Osaka)
2017-2019	MSc program, Graduate School of Agriculture, Kobe University Advisor: Hiroshi Kitagawa (Kobe University)

Awards and Honors

2023	Tadamitsu Kishimoto International Travel Award, Japanese Society for Immunology
2023	KAI 2023 Travel Award, The Korean Association of Immunologists
2023	Best Presentation Award, The 51st Annual Meeting of Japanese Society for Immunology
2022	Nissin Foods-Ando Momofuku Scholarship, Ando Sports and Food Culture Foundation
2020	JSPS Research Fellowships for Young Scientists, Japan Society for the Promotion of Science
2017	Food Sanitation Manager, Ministry of Health, Labour and Welfare
2016	Senior Laboratory Animal Technician, Japanese Society for Laboratory Animal Resources

Professional Associations

2020-Present	The Molecular Biology Society of Japan
2017-2025	Japanese Society for Immunology
2023-2024	Society for Mucosal Immunology
2023-2024	Japanese Society of Allergology
2018-2019	The Japanese Association of Anatomists
2018	The Japanese Society of Veterinary Science

Research Skills

General mice experiments
Histology / Immunohistochemistry
Flow cytometry / FACS
Cell culture (T cell / T-APC co-culture / myeloid differentiation / intestinal organoid)
CRISPR knockout experiments
Cloning
Library preparation for epigenetics (Bisulfite-seq / ATAC-seq / ChIP-seq)
Bioinformatics (Data analysis for Single-cell RNA-seq / Bulk RNA-seq / ATAC-seq / ChIP-seq)
Programming (Python / R / bash)

Bibliography

* *Senior/Corresponding Author*

Peer-reviewed Publications

2026

1 **Arai M**, Kawakami R, Nakamura Y, Naito Y, Motooka D, Sugimoto A, Kimoto T, Ohkura N, Mikami N, Sakaguchi S* (2026) Oral antigen exposure under co-stimulation blockade generates Treg cells to establish immune tolerance. *J Exp Med*. 223(3): e20251635

2025

2 Mikami N, Kawakami R, Sugimoto A, **Arai M**, Sakaguchi S* (2025) Generation of functionally stable and antigen-specific Treg cells from effector/memory T cells for cell therapy of immunological diseases. *Sci Transl Med.* 17(821): eadr6049

2024

3 Yasumizu Y, Takeuchi D, Morimoto R, Takeshima Y, Okuno T, Kinoshita M, Morita T, Kato Y, Wang M, Motooka D, Okuzaki D, Nakamura Y, Mikami N, **Arai M**, Zhang X, Kumanogoh A, Mochizuki H, Ohkura N*, Sakaguchi S* (2024) Single-cell transcriptome landscape of circulating $CD4^+$ T cell populations in autoimmune diseases. *Cell Genomics*. 4, 1000473

2023

4 Okamoto M, Sasai M, Kuratani A, Okuzaki D, **Arai M**, Wing JB, Sakaguchi S, Yamamoto M (2023) A genetic method specifically delineates Th1-type Treg cells and their roles in tumor immunity. *Cell Reports.* 42, 112813.

2022

5 **Arai M***, Fukuda A, Morimoto R, Nakamura Y, Ci Z, Sakaguchi S (2022) Protocol to evaluate cell lineage-stability of mouse natural and induced regulatory T cells using bisulfite sequencing. *STAR Protocols.* 4: 101964.

6 Yasumizu Y, Ohkura N*, Murata H, Kinoshita M, Funaki S, Nojima S, Kido K, Kohara M, Motooka D, Okuzaki D, Suganami S, Takeuchi E, Nakamura Y, Takeshima Y, **Arai M**, Tada S, Okumura M, Morii E, Shintani Y, Sakaguchi S, Okuno T, Mochizuki H (2022) Myasthenia gravis-specific aberrant neuromuscular gene expression by medullary thymic epithelial cells in thymoma. *Nat Commun.* 13: 4230.

2021

7 Kawakami R, Kitagawa Y, Chen KY, **Arai M**, Ohara D, Nakamura Y, Yasuda K, Mikami N, Lareau CA, Watanabe H, Kondo G, Hirota K, Ohkura N, Sakaguchi S* (2021) Distinct Foxp3 enhancer elements coordinate development, maintenance and function of regulatory T cells. *Immunity* 54(5): 947-961.

2020

8 **Arai M**, Mantani Y*, Nakanishi S, Haruta T, Nishida M, Yuasa H, Yokoyama T, Hoshi N, Kitagawa H (2020) Morphological and phenotypical diversity of eosinophils in the rat ileum. *Cell Tissue Res*. 381: 439-450.

2019

9 Yuasa H, Mantani Y, Miyamoto K, Nishida M, **Arai M**, Tsuruta H, Yokoyama T, Hoshi N, Kitagawa H* (2019) Effects of the expansion of bacterial colonies into the intervillous spaces on the localization of several lymphocyte lineages in the rat ileum. *J Vet Med Sci*. 81(4): 555-566.

2018

10 Masuda N, Mantani Y, Yuasa H, Yoshitomi C, **Arai M**, Nishida M, Qi WM, Kawano J, Yokoyama T, Hoshi N, Kitagawa H* (2018) Immunohistochemical study on the distribution of β-defensin 1 and β-defensin 2 throughout the respiratory tract of healthy rats. *J Vet Med Sci.* 80(3): 395-404.

11 Masuda N, Mantani Y, Yoshitomi C, Yuasa H, Nishida M, **Arai M**, Kawano J, Yokoyama T, Hoshi N, Kitagawa H* (2018) Immunohistochemical study on the secretory host defense system with lysozyme and secretory phospholipase A2 throughout rat respiratory tract. *J Vet Med Sci*. 80(2): 323-332.

2017

12 Yuasa H, Mantani Y, Masuda N, Nishida M, **Arai M**, Yokoyama T, Tsuruta H, Kawano J, Hoshi N, Kitagawa H* (2017) Mechanism of M-cell differentiation accelerated by proliferation of indigenous bacteria in rat Peyer's patches. *J Vet Med Sci*. 79(11): 1826-1835.

Conference presentations

2024

1 **Arai M**, Kawakami R, Nakamura Y, Naito Y, Motooka D, Mikami N, Sakaguchi S “Differentiation of peripherally-derived Treg cells and restoration of oral tolerance by food intake and costimulatory blockade”, *The 53th Annual Meeting of the Japanese Society for Immunology*, Nagasaki, Japan, December 2024.

2023

2 **<u>Arai M</u>**, Kawakami R, Mikami N, Nakamura Y, Sakaguchi S “Development of peripherally derived Treg cells and establishment of oral tolerance”, *The 72th Annual Meeting of the Japanese Society of Allergology*, Tokyo, Japan, October 2023.

3 **<u>Arai M</u>**, Kawakami R, Mikami N, Nakamura Y, Sakaguchi S “Antigen-specific stimulation induced and augmented peripherally derived Treg cells which establish Treg-specific epigenome and orchestrate oral tolerance”, *Keystone Symposia* “*Mechanisms of Microbiota-Immune Interactions-Towards the Next Decade”*, Utah, USA, October 2023.

4 **<u>Arai M</u>**, Kawakami R, Mikami N, Nakamura Y, Sakaguchi S “Antigen-specific stimulation induced and augmented peripherally derived Treg cells which establish Treg-specific epigenome and coordinate oral tolerance”, *Korean Association of Immunologists International Meeting 2023*, Incheon, Korea, September 2023.

5 **<u>Arai M</u>**, Sakaguchi S “Antigen-specific stimulation induced peripherally derived Treg cells which acquire Treg-specific epigenome and orchestrate oral tolerance”, *The 32th Kyoto T Cell Conference*, Kyoto, Japan, June 2023.

2022

6 **<u>Arai M</u>**, Kawakami R, Mikami N, Nakamura Y, Sakaguchi S “Antigen-specific stimulation induced peripherally derived Treg cells which establish Treg-specific epigenome and orchestrate oral tolerance”, *The 51th Annual Meeting of the Japanese Society for Immunology*, Kumamoto, Japan, December 2022.

7 **<u>Arai M</u>**, Kawakami R, Mikami N, Nakamura Y, Sakaguchi S “Antigen-specific stimulation induced peripherally derived Treg cells which establish Treg-specific epigenome and orchestrate oral tolerance”, *The 45th Annual Meeting of the Molecular Biology Society of Japan*, Chiba, Japan, November 2022.

8 **<u>Arai M</u>**, Motooka D, Sakaguchi S “Analysis of mouse small intestinal CD4+ T cells using single-cell multiplex”, *NGS EXPO 2022*, Osaka, Japan, October 2022.

2019

9 **<u>Arai M</u>**, Mantani Y, Nakanishi S, Haruta T, Nishida M, Yuasa H, Yokoyama T, Hoshi N, Kitagawa H “Diversity of eosinophil in rat small intestinal mucosa”, *The 95th Annual Meeting of Japanese Association of Anatomists*, Osaka, Japan, November 2019.

2018

10 **<u>Arai M</u>**, Mantani Y, Nishida M, Yuasa H, Yokoyama T, Hoshi N, Kitagawa H “Histological study on eosinophil lineage in rat ileal mucosa”, *Frontier Meeting for Young Scientists*, Kobe, Japan, November 2018.

11 **<u>Arai M</u>**, Mantani Y, Nishida M, Yuasa H, Yokoyama T, Hoshi N, Kitagawa H “Histological study on eosinophil lineage in rat ileal mucosa”, *The 94th Annual Meeting of Japanese Association of Anatomists*, Kobe, Japan, November 2018.

12 **<u>Arai M</u>**, Mantani Y, Nishida M, Yuasa H, Yokoyama T, Hoshi N, Kitagawa H “Immunohistochemical study on histological arrangement of immunocompetent cells with focusing on CD11b+ cells in rat small intestinal mucosa”, *The 161th meeting of the Japanese Society of Veterinary Scientists*, Tsukuba, Japan, September 2018.

Peer reviews conducted (working as co-reviewer)

Science (2), Nature Immunology (1), Nature Communications (1), Journal of Experimental Medicine (1), eLife (1)

References

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